Sky Pemberton

MA346-1

Due 10/21/24

The Data Behind Formula 1 Championship Winning Drivers

Introduction

Formula 1 is the pinnacle of motorsport, with the best drivers in the world competing to become an F1 World Champion. The sport consists of around 24 races, with points given to drivers finishing in the top 10. The driver with the most points at the end of the season becomes the World Champion. Using data from the first season in 1950 until 2022, this report discusses which factors determine an F1 driver's success in winning races and championships.

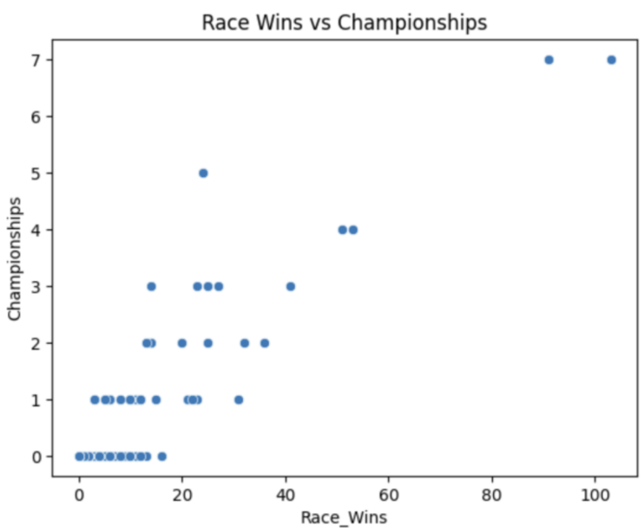
During this report, the Methods section will discuss the dataset, data cleaning, and other choices made to help provide analysis. The next section, Results, will present the findings from analysis, followed by the Discussion section which will contextualize and interpret the findings. Lastly will be the Conclusion, which will sum up the important takeaways.

Methods and Data

The dataset, from Kaggle.com, contains 868 drivers from 1950-2022 and details about their career like amount of race wins and total career points. Not a lot of data cleaning was necessary, however there were a lot of 0 values in columns such as Championships and Race Wins, which affects summary statistics. For some of the variables in figure 1, like 'Race Wins', the 25th, 50th, and 75th percentile values are all 0. This can be explained by the high driver turnover rate in F1, with 50% of drivers having less than 5 race starts, lowering averages.

	Championships	Race_Entries	Race_Starts	Pole_Positions	Race_Wins	Podiums	Fastest_Laps	Points
count	868.000000	868.000000	868.000000	868.000000	868.000000	868.000000	868.000000	868.000000
mean	0.084101	29.917051	27.694700	1.244240	1.247696	3.756912	1.261521	55.849459
std	0.524883	53.780150	52.876476	6.347512	6.491921	14.432826	5.413644	265.968614
min	0.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000
50%	0.000000	7.000000	5.000000	0.000000	0.000000	0.000000	0.000000	0.000000
75%	0.000000	29.250000	26.000000	0.000000	0.000000	0.000000	0.000000	8.000000
max	7.000000	359.000000	356.000000	103.000000	103.000000	191.000000	77.000000	4415.500000

Results



Driver	Nationality	Race_Wins	Championship Years
Keke Rosberg	Finland	5	[1982]
Mike Hawthorn	United Kingdom	3	[1958]
Phil Hill	United States	3	[1961]
Nino Farina	Italy	5	[1950]

Figure 1 Figure 2

The scatterplot in figure 1 shows a correlation between race wins and championships, with the number of championships generally increasing for drivers with more race wins. The correlation of this graph is .915. However, there were multiple drivers who won a championship with 5 race wins or less. (figure 2)

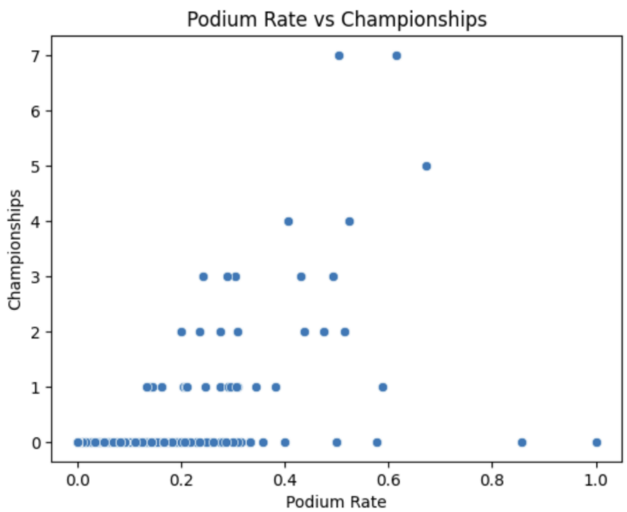


Figure 3

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Average of <=5 Win Champions, Average of all champions
0.3529297063035525 0.3374597377823853
TtestResult(statistic=0.1980314842249542, pvalue=0.8441344629596872, df=36.0)
```

Figure 4

A podium is when a driver places within the top 3 in a race, which helps the driver score more points. Podium rate is the number of podiums finishes divided by total race starts. This is also correlated with championships shown in figure 3 (.808 coefficient), with more successful drivers having higher podium rates. However, average podium rates between drivers with low race wins (35.3%) vs all other champion winning drivers (33.7%) did not show a statistically significant difference (shown in figure 4).

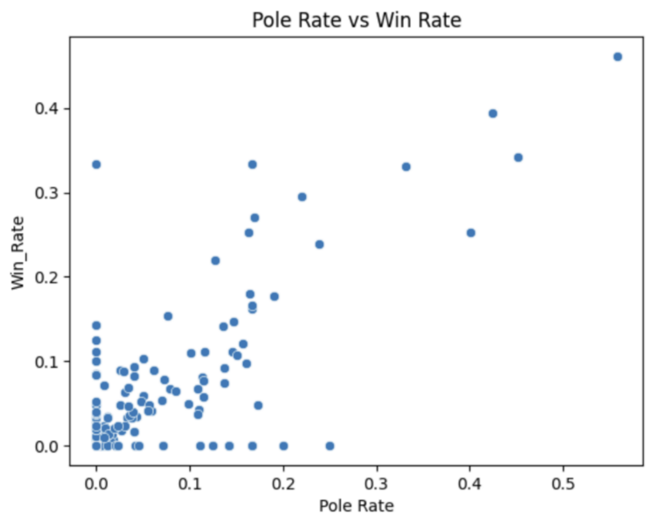
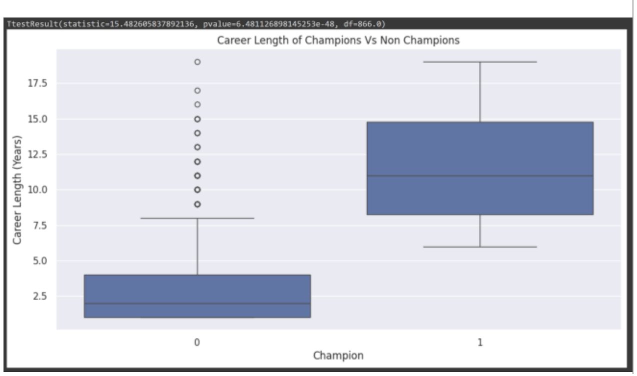


Figure 5

A ‘Pole’ is when a driver starts in the first position at the start of the race due to qualifying. Figure 5 shows a scatterplot between Pole Rate and Win Rate, attempting to explain what factors assist in race wins. The correlation of this graph is .791, showing a positive relationship.



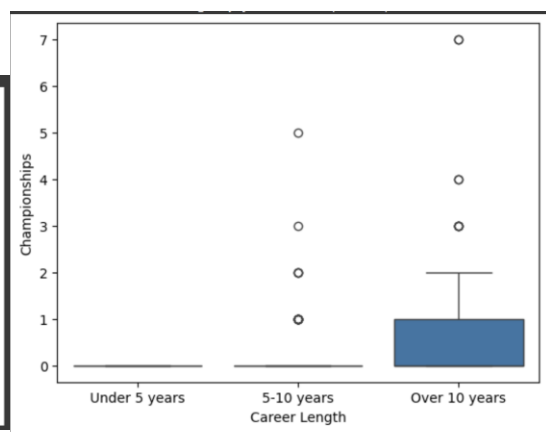


Figure 6

Figure 7

Figure 6 is a boxplot which shows the distribution of career lengths between champion and non-champion drivers. These results are statistically significant show by the p-value above the plot. Career lengths of champion drivers seem to be notably greater, however there are many outlier drivers with long careers and no championships. Additionally, figure 7 categorizes career length, showing that most championship winning drivers have careers over 10 years.

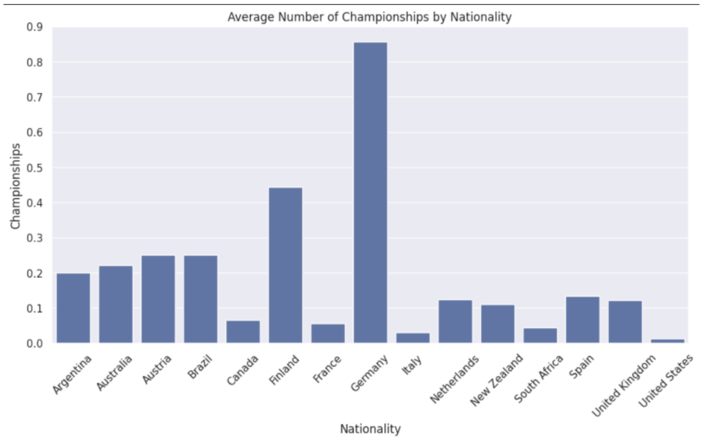


Figure 8

Figure 8 shows the average number of championships a nation has from its drivers. Germany appears to have almost 1 championship for each of its drivers, which is 12 total championships from 14 drivers.

Discussion

The number of race wins a driver has seems to be a factor in determining whether they are a championship winning driver, but there were outliers that could have potentially shown the other factors required. 4 drivers had 5 wins or less, but comparing some of their averages of Pole Rates, Podium Rates, and Points Per Race showed that they were not above average in anything. This is partially due to their seasons, 1950, 1958, 1961, having half of the races of current seasons. The drivers of these earlier seasons could be champions while only winning 1-3 races. Additionally, shown in figure 9, the careers of these drivers were shorter, leaving them with less race starts and race wins.

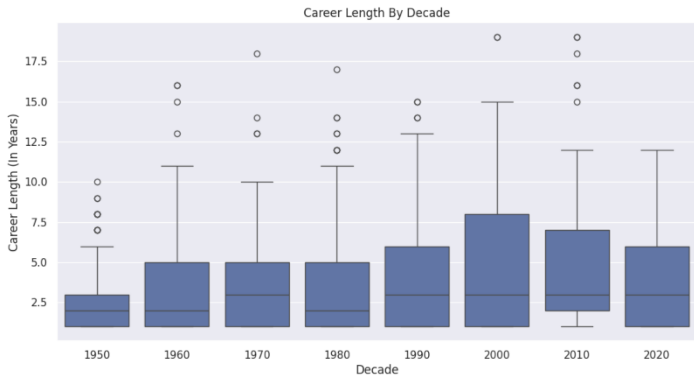


Figure 9

I decided to also check for correlation between podium rate and championships and found that drivers with higher rates tend to win more championships. This suggests that finishing within the top 3 is of similar importance, even if drivers cannot secure race wins.

In Formula 1, qualifying first (pole position) doesn't guarantee race wins, as there are many races where overtaking is common. To determine how important qualifying on pole position is for race wins, a scatterplot revealed the relationship between the variables, with a correlation coefficient of .79. Starting on pole significantly increases the likelihood of winning a race, which in turn helps drivers win championships.

Another factor that can determine a driver's success is their career length. There was found to be a significant difference in career lengths between championship winning drivers (11.7 year average career). and non-winning drivers (3.3 year average career).

Lastly, figure 8 shows which nations have the most successful drivers in winning championships. I chose to use the average number of championships, which divides the nations total championships by the number of F1 drivers to come from that nation. Germany by far has had the most success, nearly doubling Finland in second. This seemed improbable, and after looking through the data I found that German drivers before 1990 were split into separate nations, East and West Germany. This changes the results, because there are an additional 42 drivers that have not been accounted for, all with 0 championships. So, their true average since 1950 is .21, which makes Finland the most successful nation in F1, although this does not change the fact that after 1990, Germany has almost 1 championship per F1 driver.

Because Finland had the greatest number of championships per F1 driver, I researched some of their culture and learned that many children in Finland learn to drive at a young age. Driving and Racing culture is a big part of Finland, and competitive go karting is widely accessible throughout the country.

Conclusion

After analyzing factors that determine an F1 driver’s success in winning races in championships, notable correlations and relationships were found within the data. Race wins and pole positions were correlated to championship wins, however there were outliers who managed to win championships with as little as 3 race wins.

Additionally, drivers with more championships had significantly longer careers than other drivers, suggesting that experience is a necessary factor in winning championships. Lastly, there seemed to be a cultural aspect that was not quantified in the data. Finnish drivers who grew up in a racing culture have outperformed any other nation since 1950, which shows that there are other factors outside of driving performance that determine an F1 Champion.